

The Pandrosion Zeta Function and the Discriminant as Spectral Determinant

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Abstract

We introduce the *Pandrosion zeta function* $\zeta_P(s) := \sum_{k=1}^d P'(\alpha_k)^{-s}$, an analogue of the spectral zeta function where the “eigenvalues” are the derivative values $P'(\alpha_k)$ at the roots. We prove: (i) $\zeta_P(0) = d$; (ii) $\zeta_P(1) = 0$ for $d \geq 2$ (the *vanishing identity*, proved via partial fractions); (iii) a hierarchy of $d - 1$ vanishing relations $\sum \alpha_k^m / P'(\alpha_k) = 0$ for $m \leq d - 2$; and (iv) the discriminant is the spectral determinant: $\exp(-\zeta'_P(0)) = \prod P'(\alpha_k) = \pm D(P)^2$. This gives a spectral-analytic formulation of all results from Papers 20–22.

1 Definition and basic properties

Definition 1.1 (Pandrosion zeta function). For $P(z) = \prod_{k=1}^d (z - \alpha_k)$ with simple roots:

$$\zeta_P(s) := \sum_{k=1}^d P'(\alpha_k)^{-s}, \quad (1)$$

where $P'(\alpha_k) = Q(\alpha_k, \alpha_k) = \prod_{j \neq k} (\alpha_k - \alpha_j)$ is the Pandrosion self-value (Paper 22).

Proposition 1.2 (Trivial value). $\zeta_P(0) = d$.

2 The vanishing identity

Theorem 2.1 (Pandrosion vanishing). For $d \geq 2$:

$$\boxed{\zeta_P(1) = \sum_{k=1}^d \frac{1}{P'(\alpha_k)} = 0.} \quad (2)$$

Proof. The partial fraction decomposition of $1/P(z)$ is:

$$\frac{1}{P(z)} = \sum_{k=1}^d \frac{1}{P'(\alpha_k)(z - \alpha_k)}. \quad (3)$$

As $z \rightarrow \infty$:

$$\frac{1}{P(z)} = z^{-d}(1 + O(z^{-1})) \quad (\text{no } z^{-1} \text{ term for } d \geq 2).$$

The partial fraction sum expands as:

$$\sum_k \frac{1}{P'(\alpha_k)(z - \alpha_k)} = \frac{1}{z} \sum_k \frac{1}{P'(\alpha_k)} + O(z^{-2}).$$

Matching the z^{-1} coefficient: $\sum 1/P'(\alpha_k) = 0$. □

Theorem 2.2 (Higher vanishing relations).

$$\sum_{k=1}^d \frac{\alpha_k^m}{P'(\alpha_k)} = \begin{cases} 0 & \text{if } 0 \leq m \leq d-2, \\ 1 & \text{if } m = d-1. \end{cases} \quad (4)$$

Proof. The coefficient of z^{-1} in $z^m/P(z) = z^{m-d}(1 + c_1 z^{-1} + \dots)$ is 0 for $m \leq d-2$ and 1 for $m = d-1$. By partial fractions:

$$\frac{z^m}{P(z)} = \sum_k \frac{\alpha_k^m}{P'(\alpha_k)(z - \alpha_k)} + \text{polynomial in } z.$$

The coefficient of z^{-1} on the right is $\sum \alpha_k^m / P'(\alpha_k)$. □

Remark 2.3. The $d-1$ vanishing relations form a *biorthogonality system*: the vectors $(\alpha_1^m, \dots, \alpha_d^m)$ for $m = 0, \dots, d-2$ are orthogonal to the vector $(1/P'(\alpha_1), \dots, 1/P'(\alpha_d))$. The normalisation at $m = d-1$ fixes the scale.

3 The discriminant as spectral determinant

Theorem 3.1 (Spectral determinant).

$$\boxed{\exp(-\zeta'_P(0)) = \prod_{k=1}^d P'(\alpha_k) = (-1)^{d(d-1)/2} D(P)^2,} \quad (5)$$

where $D(P) = \prod_{i < j} (\alpha_i - \alpha_j)$ is the discriminant.

Proof. $\zeta'_P(s) = -\sum \log P'(\alpha_k) \cdot P'(\alpha_k)^{-s}$, so $\zeta'_P(0) = -\sum \log P'(\alpha_k) = -\log \prod P'(\alpha_k)$. Exponentiating: $\exp(-\zeta'_P(0)) = \prod P'(\alpha_k)$.

By Paper 22: $P'(\alpha_k) = \prod_{j \neq k} (\alpha_k - \alpha_j)$, so $\prod_k P'(\alpha_k) = \prod_k \prod_{j \neq k} (\alpha_k - \alpha_j) = (-1)^{d(d-1)/2} D(P)^2$. □

This is the polynomial analogue of the Minakshisundaram–Pleijel zeta function, where the spectral determinant of a Laplacian on a manifold is computed via $\exp(-\zeta'(0))$. Here, the “manifold” is the root configuration of P , and the “Laplacian eigenvalues” are $P'(\alpha_k)$.

4 The energy zeta function

Definition 4.1. The **energy zeta function** is:

$$Z_P(s) := \sum_{k=1}^d E_P(\alpha_k)^{-s} = \sum_{k=1}^d P'(\alpha_k)^{-2s} = \zeta_P(2s). \quad (6)$$

s	$Z_P(s)$	Interpretation
-1	$\sum E_P(\alpha_k)$	Total self-energy
$-1/2$	$\sum P'(\alpha_k) $	Total field strength
0	d	Degree
$1/2$	$\sum 1/ P'(\alpha_k) $	Sum of residue norms
1	$\sum 1/E_P(\alpha_k)$	“Capacitance”

5 Analogy with the Riemann zeta function

Property	Riemann ζ	Pandrosion ζ_P
Definition	$\sum n^{-s}$	$\sum P'(\alpha_k)^{-s}$
Value at 0	$-1/2$	d
Vanishing?	$\zeta(-2n) = 0$ (trivial zeros)	$\zeta_P(1) = 0$
Derivative at 0	$-\log \sqrt{2\pi}$	$-\log D(P) ^2$
Spectral determinant	$\sqrt{2\pi}$	$ D(P) ^2$
Euler product	$\prod (1 - p^{-s})^{-1}$	$\prod (1 - r_k)$ (Paper 25)

The discriminant $D(P)$ is the Pandrosion analogue of $\sqrt{2\pi}$: the *regularised volume* of the root configuration.

6 Conclusion

The Pandrosion zeta function $\zeta_P(s) = \sum P'(\alpha_k)^{-s}$ encodes the entire spectral information of the root system in a single meromorphic function of s . Its key features are:

1. **Vanishing at $s = 1$:** $\sum 1/P'(\alpha_k) = 0$, a fundamental constraint on derivative values at roots.
2. **Higher vanishing:** $d - 1$ biorthogonality relations from partial fractions.
3. **Spectral determinant:** $\exp(-\zeta'_P(0)) = \pm D(P)^2$, giving the discriminant a zeta-theoretic meaning.

Together with Paper 22 ($D^2 = \prod E_P(\alpha_k)$) and Paper 26 (electrostatic interpretation), this completes the analytic picture of the Pandrosion root spectrum.

References

- [1] S. Minakshisundaram, Å. Pleijel, *Some properties of the eigenfunctions of the Laplace-operator*, Canad. J. Math. **1** (1949), 242–256.
- [2] I. Besevic, *The Pandrosion discriminant*, Preprint (2026).